

# SQL Subqueries easy level

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## Tables

```
CREATE TABLE employees (  
    id INT,  
    name VARCHAR(50),  
    salary INT,  
    dept_id INT  
);
```

```
CREATE TABLE departments (  
    dept_id INT,  
    dept_name VARCHAR(50)  
);
```

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## Sample Data

```
INSERT INTO employees VALUES  
(1, 'Amit', 30000, 1),  
(2, 'Rahul', 40000, 2),  
(3, 'Priya', 35000, 1),  
(4, 'Neha', 45000, 3),  
(5, 'Karan', 25000, 2),  
(6, 'Anjali', 50000, 3);
```

```
INSERT INTO departments VALUES  
(1, 'HR'),  
(2, 'IT'),  
(3, 'Finance');
```

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# Subquery Questions with Solutions

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## ◆ Basic Subqueries

### 1. Employees earning more than average salary

```
SELECT * FROM employees
WHERE salary > (SELECT AVG(salary) FROM employees);
```

### 2. Employees earning less than average salary

```
SELECT * FROM employees
WHERE salary < (SELECT AVG(salary) FROM employees);
```

### 3. Employee with maximum salary

```
SELECT * FROM employees
WHERE salary = (SELECT MAX(salary) FROM employees);
```

### 4. Employee with minimum salary

```
SELECT * FROM employees
WHERE salary = (SELECT MIN(salary) FROM employees);
```

### 5. Employees earning more than Rahul

```
SELECT * FROM employees
WHERE salary > (SELECT salary FROM employees WHERE name='Rahul');
```

### 6. Employees in same department as Amit

```
SELECT * FROM employees
WHERE dept_id = (SELECT dept_id FROM employees WHERE name='Amit');
```

### 7. Employees with minimum salary

```
SELECT * FROM employees
```

```
WHERE salary = (SELECT MIN(salary) FROM employees);
```

### **8. Employees with maximum salary**

```
SELECT * FROM employees  
WHERE salary = (SELECT MAX(salary) FROM employees);
```

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## **◆ IN Subqueries**

### **9. Employees whose dept\_id exists in departments**

```
SELECT * FROM employees  
WHERE dept_id IN (SELECT dept_id FROM departments);
```

### **10. Employees in HR**

```
SELECT * FROM employees  
WHERE dept_id IN (  
    SELECT dept_id FROM departments WHERE dept_name='HR'  
);
```

### **11. Employees in IT**

```
SELECT * FROM employees  
WHERE dept_id IN (  
    SELECT dept_id FROM departments WHERE dept_name='IT'  
);
```

### **12. Employees in Finance**

```
SELECT * FROM employees  
WHERE dept_id IN (  
    SELECT dept_id FROM departments WHERE dept_name='Finance'  
);
```

### **13. Employees in departments with dept\_id > 1**

```
SELECT * FROM employees  
WHERE dept_id IN (  
    SELECT dept_id FROM departments WHERE dept_id > 1
```

);

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## ◆ NOT IN Subqueries

### 14. Employees NOT in HR

```
SELECT * FROM employees
WHERE dept_id NOT IN (
    SELECT dept_id FROM departments WHERE dept_name='HR'
);
```

### 15. Employees NOT in IT

```
SELECT * FROM employees
WHERE dept_id NOT IN (
    SELECT dept_id FROM departments WHERE dept_name='IT'
);
```

### 16. Employees NOT in dept\_id = 3

```
SELECT * FROM employees
WHERE dept_id NOT IN (
    SELECT dept_id FROM departments WHERE dept_id=3
);
```

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## ◆ Scalar Subqueries

### 17. Salary > average salary of dept 1

```
SELECT * FROM employees
WHERE salary > (
    SELECT AVG(salary) FROM employees WHERE dept_id=1
);
```

### 18. Salary < max salary of dept 2

```
SELECT * FROM employees
WHERE salary < (
    SELECT MAX(salary) FROM employees WHERE dept_id=2
);
```

```
);
```

### **19. Salary = min salary of dept 3**

```
SELECT * FROM employees
WHERE salary = (
    SELECT MIN(salary) FROM employees WHERE dept_id=3
);
```

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## **◆ EXISTS Subqueries**

### **20. Employees where department exists**

```
SELECT * FROM employees e
WHERE EXISTS (
    SELECT 1 FROM departments d WHERE e.dept_id = d.dept_id
);
```

### **21. Employees where department does NOT exist**

```
SELECT * FROM employees e
WHERE NOT EXISTS (
    SELECT 1 FROM departments d WHERE e.dept_id = d.dept_id
);
```

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## **◆ Nested Subqueries**

### **22. Salary > average salary of IT**

```
SELECT * FROM employees
WHERE salary > (
    SELECT AVG(salary) FROM employees
    WHERE dept_id = (
        SELECT dept_id FROM departments WHERE dept_name='IT'
    )
);
```

### **23. Salary < average salary of HR**

```
SELECT * FROM employees
WHERE salary < (
    SELECT AVG(salary) FROM employees
    WHERE dept_id = (
        SELECT dept_id FROM departments WHERE dept_name='HR'
    )
);
```

#### **24. Salary > min salary of Finance**

```
SELECT * FROM employees
WHERE salary > (
    SELECT MIN(salary) FROM employees
    WHERE dept_id = (
        SELECT dept_id FROM departments WHERE dept_name='Finance'
    )
);
```

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### **◆ Name-based Practice**

#### **25. Salary > Priya**

```
SELECT * FROM employees
WHERE salary > (
    SELECT salary FROM employees WHERE name='Priya'
);
```

#### **26. Salary < Neha**

```
SELECT * FROM employees
WHERE salary < (
    SELECT salary FROM employees WHERE name='Neha'
);
```

#### **27. Same department as Karan**

```
SELECT * FROM employees
WHERE dept_id = (
    SELECT dept_id FROM employees WHERE name='Karan'
);
```

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## ◆ Bonus

### 28. Second highest salary

```
SELECT MAX(salary) FROM employees
WHERE salary < (SELECT MAX(salary) FROM employees);
```

### 29. Salary > avg and < max

```
SELECT * FROM employees
WHERE salary > (SELECT AVG(salary) FROM employees)
AND salary < (SELECT MAX(salary) FROM employees);
```

### 30. Employees in department with highest average salary

```
SELECT * FROM employees
WHERE dept_id = (
    SELECT dept_id FROM employees
    GROUP BY dept_id
    ORDER BY AVG(salary) DESC
    LIMIT 1
);
```